

C47 celestial navigation programs

User manual — how each program works, with worked results

These programs give the data needed for sight reduction (GHA, SHA, declination, Hc, Zn) and for planning (sunrise, sunset, twilight, Moon phase, sky picture). They support, and do not replace, the Nautical Almanac. All results are apparent geocentric positions, as in the almanac. The time argument is UT (UT1): for a sight timed in UTC, add DUT1 (under 0.9 s).

Overview

Program	What it gives	Input	Output	Needs
SUNA	Sun and Aries	X = JD	X GHA [⊙] , Y Dec [⊙] , Z GHA ^Υ	matrices from MATA
MATA	builds Earth-orbit and nutation matrices	—	VL, VB, VR, NU	run once
STAR	57 navigation stars + Polaris	Y = JD, X = star no.	X GHA, Y Dec, Z SHA	SUNA, matrix ST
MATST	builds star catalogue matrix	—	ST	run once
SNAM	star name	X = star no.	name on screen	—
SUNRISE	rise, set, twilight, meridian passage	Z = JD 0h, Y = lat, X = lon	X = UT hours	SUNA
PHAS	Moon phase	X = JD	X % illuminated, Y age (days)	SUNA
HORZ	picture of the sky on the horizon	Z = JD, Y = lat, X = lon	screen drawing	SUNA, STAR
ALM	Sun, Aries, Moon, planets from tables	X = t (hours in block)	X GHA, Y Dec (HP)	coefficients from the tables
CHZ	Hc/Zn and the inverse	T lat, Z lon, Y Dec, X GHA	X Hc, Y Zn	—
SUNSD	Sun semi-diameter	X = JD	X = SD (')	SUNA
HALMV	sight-planning screen: chart + data	Z = JD, Y = lat, X = lon	screen	see HALMV section
ALMF	full-page almanac screen	Z = JD, Y = lat, X = lon	screen	see ALMF section
PTXB / PTXT	pixel text (big / small font)	Z = y, Y = x, X = text	drawing	—

Conventions

- **Julian Date (JD):** use the C47 date function (Date→J) and add UT/24. It must be a plain number: for example 23 Sep 2026 12:00 UT = **2461307**, and 0h UT that day = 2461306.5.
- **Signs:** latitude N +, S −; longitude E +, W −; declination negative = South.
- **Angles** come out in decimal degrees. Use →HMS or the d.mmm display to read degrees and minutes.
- **Setup order:** load SUNA, MATA, STAR, MATST, CHZ, SUNRISE, PHAS, SNMU, PTXB (and PTXT), then the screens HALMV, ALMF, HALM, HORZ. XEQ MATA; XEQ MATST. MATA and MATST can then be deleted; never delete the matrices. Then save the whole state (I/O menu, SAVEST) so one LOADST restores everything.

SUNA — Sun and Aries

Computes the Sun's GHA and declination and the GHA of Aries for any instant, with no tables. It is also the engine used by STAR, SUNRISE, PHAS and HORZ.

How it works

- Time: T and τ (Julian centuries and millennia of TT since J2000; TT = UT + 69.2 s).
- Earth's heliocentric longitude L, latitude B and distance R from the VSOP87D series stored in VL, VB, VR (56 + 3 + 7 terms). The loops count rows with DSE 55.

- Sun = L + 180°, β = −B; FK5 correction.
- Nutation Δψ, Δε from the 10 largest IAU 1980 terms (matrix NU). The Moon arguments D, M, M', F are kept in R60–R63 for PHAS.
- Apparent longitude λ = Sun + Δψ − 20.4898"/R (aberration). Obliquity ε = 84381.448" − 46.815"·T + Δε.
- Dec = asin(sin β cos ε + cos β sin ε sin λ); RA from →POL.
- GHA^Υ = GMST + Δψ·cos ε; GHA[⊙] = GHA^Υ − RA (mod 360).

Inputs and outputs

Input	X = JD (UT1)
Output	X = GHA [⊙] , Y = Dec [⊙] , Z = GHA ^Υ (degrees). Also R73 = Sun distance R (au): SD [⊙] = 15.99' ÷ R73.
Registers	R50–R55, R60–R67, R70–R81

Example with results

23 Sep 2026, 12:00:00 UT: key **2461307 XEQ SUNA**

	Result	Deg / min	Almanac page
X GHA [⊙]	1.9069849	1° 54.4'	1° 54.4'
Y Dec [⊙]	-0.1931700	S 0° 11.6'	S 0° 11.6'
Z GHA ^Υ	182.3525029	182° 21.2'	182° 21.2'
R73 → SD [⊙]	1.003445 au	15.94'	15.94'

Notes

- Accuracy vs JPL/IAU reference 2000–2050: GHA ≤ 0.016', Dec ≤ 0.007', GHA^Υ ≤ 0.003'. Verified on the C47.
- Coefficients never need updating. ΔT (69.2 s, line 4) affects the Sun by only 0.0007' per second.

MATA and MATST — setup programs

They contain no calculation. Each creates matrices and fills them with coefficients, so you don't have to type about 590 numbers by hand. Run each once; afterwards the programs can be deleted.

Program	Creates	Contents
MATA	VL 56×4, VB 3×4, VR 7×4	VSOP87D Earth series rows: k, A, B, C (term = A·cos(B + C·τ)·τ ^k)
MATA	NU 10×9	IAU 1980 nutation: multipliers of D, M, M', F, Ω and coefficients S, S', C, C'
MATST	ST 58×4	Star J2000 RA, Dec (deg) and proper motions (mas/yr), stars 1–57 in almanac order, 58 = Polaris

Memory kept afterwards: about 5.7 KB (VL, VB, VR, NU) + 3.7 KB (ST).

STAR — navigation stars

Gives GHA, SHA and declination of any of the 57 almanac navigation stars or Polaris (58).

How it works

- Runs SUNA for the instant (nutation, obliquity, Sun longitude, GHA^Υ). Entry STR2 skips this when SUNA has already run (used by HORZ).
- Reads the star's J2000 position and proper motion from ST and applies proper motion.
- Precession J2000 → date with the IAU 2006 angles ζ, z, θ.
- Converts to ecliptic coordinates and adds nutation Δψ and annual aberration there (accurate even for Polaris), then back with the true obliquity.
- SHA = 360° − RA; GHA = GHA^Υ + SHA.

Inputs and outputs

Input	Y = JD (UT1), X = star number 1-58
Output	X = GHA, Y = Dec, Z = SHA
Registers	R56–R59, R68–R69, R82–R89 (plus SUNA's)

Example with results

23 Sep 2026, 12:00 UT: key **2461307 ENTER 18 XEQ STAR** (Sirius), and 58 for Polaris

Star	GHA	SHA	Dec
18 Sirius	80° 46.2′	258° 25.1′	S 16° 44.9′
58 Polaris	135° 19.7′	312° 58.5′	N 89° 22.4′

Notes

- Accuracy 2025–2050: 57 stars $\leq 0.003'$; Polaris $\leq 0.04'$ in hour angle (tiny on the sky). Checked against USNO for 30 stars: $\leq 0.009'$.
- Star numbers are the official almanac numbers; the list in STAR_list.txt gives them by number and alphabetically.

SNAM — star names

Shows the name of a star from its number: key **18 XEQ SNAM** → display "18 Sirius". It stores the number in R49, jumps with GTO IND 49 to local label 01–58, loads the name as text and shows it with AVIEW. The stack is unchanged.

To show the name automatically after STAR, add RCL 82 · XEQ "SNAM" · R↓ before STAR's last RTN.

SUNRISE — rise, set, twilight, meridian passage

Times of sunrise, sunset, civil and nautical twilight and meridian passage for any position, directly in UT.

How it works

- Starts from a first guess (06h for morning events, 18h for evening, 12h for transit).
- Runs SUNA at that time, computes the hour angle at which the Sun reaches the event altitude: $\cos H = (\sin h_0 - \sin \phi \sin \delta) / (\cos \phi \cos \delta)$.
- Corrects the time by $(\text{target LHA} - \text{actual LHA}) / 15$ and repeats until the correction is under 0.04 s (3–4 passes).
- Event altitudes h_0 : rise/set $-0^\circ 50'$ (upper limb, refraction), civil -6° , nautical -12° . If the Sun never reaches it, the result is 99.

Inputs and outputs

Input	Z = JD at 0h UT of the day, Y = latitude (N+), X = longitude (E+, W–)
Labels	NTWA / CTWA (morning twilight), RISE, TRAN, SET, CTWP / NTWP (evening twilight)
Output	X = UT in decimal hours (→HMS to read); 99 = no event
Registers	R90–R99

Example with results

23 Sep 2026, latitude 10° N, longitude 0° : key **2461306.5 ENTER 10 ENTER 0**, then the label

Label	Event	UT	Almanac page (10° N)
NTWA	nautical twilight am	05:03:47	05 04
CTWA	civil twilight am	05:28:09	05 28
RISE	sunrise	05:49:09	05 49
TRAN	meridian passage	11:52:22	11 52 22
SET	sunset	17:55:28	17 56
CTWP	civil twilight pm	18:16:27	18 17
NTWP	nautical twilight pm	18:40:49	18 41

Medellín (6.25° N, 75.57° W): sunrise 10:51:24 UT (05:51 local), sunset 22:57:41 UT (17:57 local). 89.5° N: RISE gives 99 (no sunrise).

Notes

- Checked against an exact search on the JPL/IAU reference: agrees to the second.
- The almanac holds the noon declination fixed for the whole day, so its times can differ by up to a minute (sunset and civil pm above). The program is the more exact one.

PHAS — Moon phase

Percentage of the Moon's disk illuminated and the age of the Moon (days since new moon), as printed in the almanac.

How it works

- Runs SUNA, which leaves the mean Moon arguments D, M, M', F in R60–R63.
- Phase angle i (Meeus 48.4) $\rightarrow$ illuminated fraction $k = (1 + \cos i)/2$.
- Age: days since the mean new moon, corrected to the true new moon with Meeus ch. 49 terms; near new moon it checks the neighbouring lunation.

Inputs and outputs

Input	X = JD (UT1)
Output	X = % illuminated, Y = age (days)
Registers	R41–R48

Example with results

23 Sep 2026, 12:00 UT: key **2461307 XEQ PHAS** $\rightarrow$ X = **89.1 %** (waxing gibbous), Y = **12.36 days**. 10 Oct 2026 12h: 0.0 %, 29.36 d (new moon that evening); 11 Oct 12h: 0.8 %, 0.84 d.

Notes

- Accuracy vs JPL: illumination ≤ 0.3 %, age ≤ 2 minutes.

HORZ — sky picture on the horizon

Draws a chart of the whole sky: azimuth Zn across (N–E–S–W–N, ticks every 30°), altitude Hc up (ticks and labels at 30°, 60°, 90°), the celestial equator dotted, the Sun (☉) and every navigation star above the horizon (★ with its number).

How it works

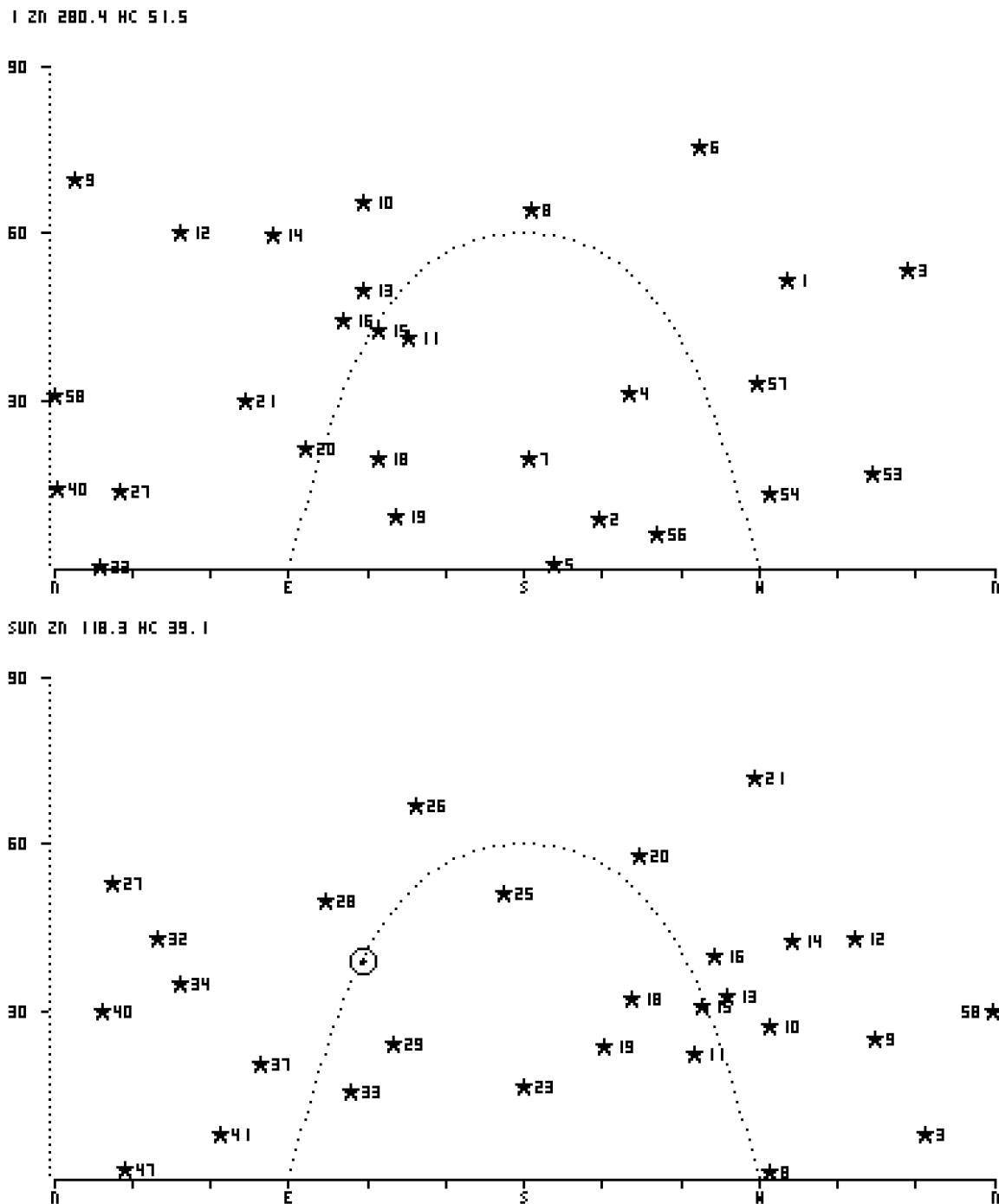
- Runs SUNA once, then STAR through entry STR2 for stars 1–58 (fast: no repeated SUNA). The celestial equator is drawn as dots every 2° of hour angle.
- For each object: $LHA = GHA + \lambda$; $Hc = \arcsin(\sin \phi \sin \delta + \cos \phi \cos \delta \cos LHA)$; Zn from $\rightarrow$ POL of $(-\cos \delta \sin LHA, \cos \phi \sin \delta - \sin \phi \cos \delta \cos LHA)$.
- Screen: column = $20 + Zn \times 375/360$; row = $225 - Hc \times 200/90$ (from the top; converted for the C47, which counts rows from the bottom). Only objects with $Hc > 0$ are drawn.
- Text (N E S W, 30 60 90, star numbers, the info line) is drawn with PTXT, the small AGRAPH font (PTNS for integers, PT1 for one decimal); the star and sun symbols with PTXB (* and @). Lines and dots use PHL and PIXEL.

Inputs and outputs

Input	Z = JD (UT1), Y = latitude (N+), X = longitude (E+)
Output	drawing on the 400×240 screen; R96 = Hc and R97 = Zn of the last object
Needs	SUNA, STAR (STR2), CHZ, PTXT, PTXB
Registers	R90–R99, R36, R37, R42, R44, R86, R93

Example with results

23 Sep 2026, 30° N 0° E. Night 03:00 UT: key **2461306.625 ENTER 30 ENTER 0 XEQ HORZ**; day 09:00 UT: 2461306.875. Simulated screens:



Notes

- Hc and Zn agree with the reference within 0.0013'; Sun Hc/Zn match USNO (88.0833° / 264.215° vs 88.0834° / 264.214°).
- Stars are drawn even in daylight; they are only observable in twilight.

HALMV — sight-planning screen (chart + almanac data)

One screen for twilight star sights: on the left a picture of the sky (Hc up, Zn across) with the Sun, Moon, planets and the chosen stars; on the right the date and time, your DR, GHA Aries, the Sun's GHA, Dec and SD, a table of Hc and Zn (Sun, then the Moon and planets that are above the horizon, then the brightest stars, 10 rows), the day's sun times, and the Moon's phase, HP and SD.

How it works

- First computes the sun times with SUNRISE (NTWA, RISE, TRAN, SET, NTWP for the UT date of the JD) and the Moon with PHAS, and keeps them in R13–R19.

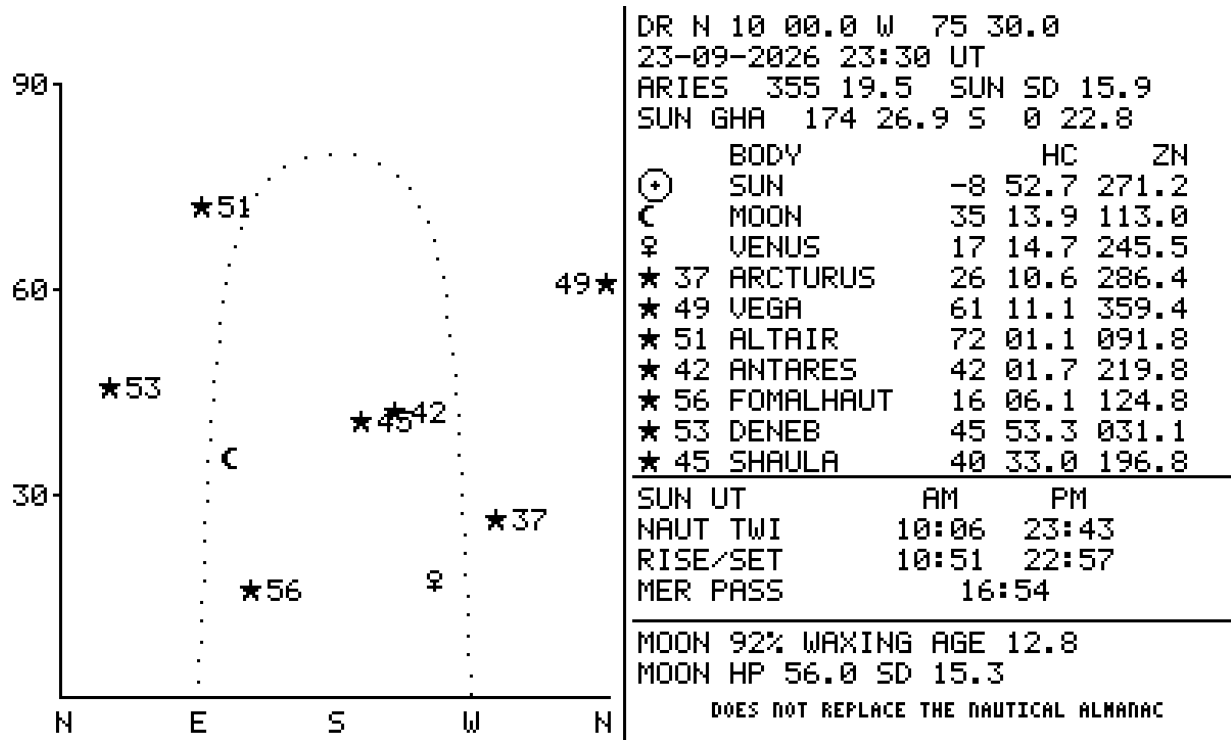
- Runs SUNA once (Sun GHA, Dec, GHA Aries; SD = 15.994' / R from R73), then STAR through entry STR2 for stars 1-58.
- Moon (MOO2) and planets (PLN2) are computed without repeating SUNA. Symbols: ☾ Moon, ♀ Venus, ♂ Mars, ♃ Jupiter, ♄ Saturn. Stars: in order of brightness (SBRT), only those higher than 10°, until the table has 10 rows.
- Hc and Zn: CHZ subroutine HCZ. Chart: column = 18 + Zn × 178/360, row = 14 + Hc × 200/90; the celestial equator is dotted every 3° of hour angle. South latitude: chart centred on N (S W N E S).
- All text and numbers are drawn with PIXEL through PTXB (5×7 font): strings with the native C47 string commands, numbers with PINB, PDM (deg + min), PZN (bearing), PHM (hh:mm), PF1 and PDAT (date from JD). Star names from SNMU. The screen is held with PAUSE.

Inputs and outputs

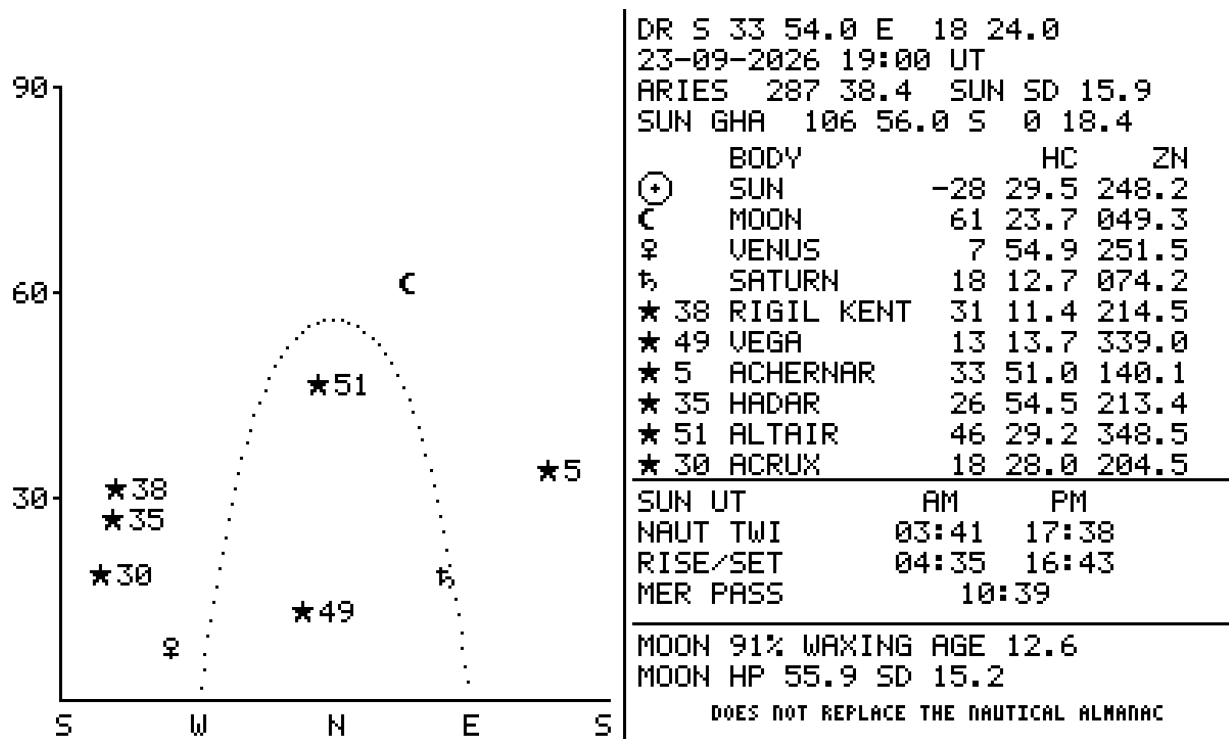
Input	Z = JD (UT1), Y = latitude (N+), X = longitude (E+)
Screen	left: chart; right: DR, date + UT, Aries, Sun SD, Sun GHA/Dec, table BODY / HC / ZN, naut. twilight AM/PM, rise/set, mer pass (UT), Moon % + waxing/waning + age
Needs	SUNA, STAR, CHZ, SUNRISE, PHAS, MOON, PLAN, SBRT, SNMU, PTXB + all matrices (MATA, MATST, MATM, MATP)
Registers	R10-R29, R37, R40-R48 (+ those of the programs it calls)
Time	about 110 000 steps

Example with results

23 Sep 2026, 23:30 UT, DR 10° 00' N 075° 30' W: key **2461307.4792 ENTER 10 ENTER -75.5 XEQ HALMV**. Simulated screen:



Southern example, 33° 54' S 018° 24' E, 19:00 UT (chart centred on N):



Notes

- Sun times checked against the JPL-based reference for this position: naut. twilight 10:05:46 / 23:42:41, rise 10:51:08, set 22:57:20, mer pass 16:54:18 UT — the screen rounds to the minute as the almanac does.
- All times are UT for the UT date of the JD, at the DR entered; sea level, standard refraction (as the almanac). West of Greenwich an evening event after 24h UT shows as 00:xx (next UT day).
- Hc and Zn are computed from the DR: use them to identify stars and pre-set the sextant. Near the chart edges two star numbers may overlap; the table is always exact.
- Sun SD = $959.63''/R$; it can differ by 0.1' from the printed page, which gives one SD per three days.

ALMF — full-page almanac screen

The same data as HALMV without the chart, laid out like an almanac page: GHA and Dec as well as Hc and Zn, with the same 10-row rule as HALMV (Sun, then the Moon and planets above the horizon, then the brightest stars higher than 10°), the sun times, Moon phase, Moon HP and SD, and Sun SD. The last line reads DOES NOT REPLACE THE NAUTICAL ALMANAC (also on HALMV).

How it works

- Same calculations and table rule as HALMV.
- Top line: date, UT, DR and GHA Aries. Table: symbol, number, name, GHA, Dec, Hc, Zn. Bottom left: nautical twilight and rise/set in AM/PM columns and meridian passage; bottom right: Moon % and waxing/waning, age, Sun SD.

Inputs and outputs

Input	Z = JD (UT1), Y = latitude (N+), X = longitude (E+)
Needs	same programs as HALMV
Registers	same as HALMV
Time	about 100 000 steps

Example with results

Same instant and DR as the HALMV example: key **2461307.4792 ENTER 10 ENTER -75.5 XEQ ALMF**. Simulated screen:

23-09-2026 23:30 UT DR N 10 00.0 W 75 30.0 ARIES 355 19.5

	BODY	GHA	DEC	HC	ZN
⊕	SUN	174 26.9	S 0 22.8	-8 52.7	271.2
☾	MOON	25 09.3	S 12 19.9	35 13.9	113.0
♀	VENUS	142 57.4	S 19 47.4	17 14.7	245.5
★ 37	ARCTURUS	141 06.4	N 19 02.7	26 10.6	286.4
★ 49	VEGA	75 51.7	N 38 48.8	61 11.1	359.4
★ 51	ALTAIR	57 17.9	N 8 56.5	72 01.1	091.8
★ 42	ANTARES	107 33.7	S 26 29.5	42 01.7	219.8
★ 56	FOMALHAUT	10 32.1	S 29 28.7	16 06.1	124.8
★ 53	DENEK	44 44.1	N 45 22.8	45 53.3	031.1
★ 45	SHAULA	91 28.0	S 37 07.5	40 33.0	196.8

SUN UT	AM	PM	MOON 92%	WAXING
NAUT TWI	10:06	23:43	AGE 12.8	DAYS
RISE/SET	10:51	22:57	SUN SD 15.9	
MER PASS	16:54		MOON HP 56.0	SD 15.3

DOES NOT REPLACE THE NAUTICAL ALMANAC

Notes

- GHA and Dec are the almanac values for that instant (Sun $\leq 0.02'$, stars $\leq 0.003'$ against JPL); read them directly for sight reduction.
- HALM is a third layout of the same data: chart on the top half, table (GHA, Dec, Hc, Zn) on the bottom half.

Supporting programs for the screens

Program	Use	Stack
PTXB	big 5x7 pixel font; * = star, @ = sun, also % ; entries PINB, PDM, PZN, PHM, PF1, PDAT	Z = y, Y = x, X = text or number $\rightarrow$ Y = y, X = next x
PTXT	small 3x5 pixel font, same characters and stack	as PTXB
SNMU	star name in capitals, returned in X (no display)	X = star no. $\rightarrow$ X = name
SUNSD	Sun semi-diameter 15.994' / R	X = JD $\rightarrow$ X = SD (')
CHZ / HCZ / DHA	Hc and Zn; inverse gives Dec, LHA, GHA	CHZ: T lat, Z lon, Y Dec, X GHA $\rightarrow$ X Hc, Y Zn

ALM — table method (Sun, Aries, Moon, planets)

Evaluates short Chebyshev polynomials whose coefficients you key in from the tables in C47_GHA_Dec_Almanac.pdf or the Excel file. It is the program for the Moon and the planets (Venus, Mars, Jupiter, Saturn).

How it works

- Each table line covers a block: 8 days (192 h) for Sun, Aries and planets; 1 day (24 h) for the Moon.
- $x = 2t/\text{span} - 1$, where t = hours since the start of the block.
- CHEB sums the series with the Clenshaw recurrence; ALM does it for GHA (R10...) and Dec (R20...); HP does the Moon's horizontal parallax (R30-R33).

Inputs and outputs

Store	R01 = n (5, Moon 6), R02 = span (192, Moon 24), GHA coefficients R10..., Dec R20..., Moon HP R30-R33
Input	X = t (hours since the block start, UT1)
Output	X = GHA, Y = Dec; XEQ HP $\rightarrow$ horizontal parallax (arcmin)

Store **R01 = n (5, Moon 6), R02 = span (192, Moon 24), GHA coefficients R10..., Dec R20..., Moon HP R30-R33**

Registers R01-R09, R10-R33, R40-R41

Example with results

Sun, 23 Sep 2026 12:00 UT: block 2026-09-17, $t = 6 \times 24 + 12 = 156$

	c0	c1	c2	c3	c4
GHA	1621.68475	1440.35480	-0.00108	-0.00024	-0.00000
Dec	0.77751	-1.55320	-0.00149	0.00024	-0.00000

Result: GHA 1° 54.4', Dec S 0° 11.6' (same as SUNA).

Moon, 23 Sep 2026 19:00 UT: block 2026-09-23, $t = 19$

	c0	c1	c2	c3	c4	c5
GHA	217.86332	174.54982	0.01758	-0.00081	-0.00002	0.00000
Dec	-14.82363	2.56120	0.04565	-0.00100	-0.00001	-0.00000
HP	55.738	0.303	0.004	-0.000		

Result: GHA 319° 40.8', Dec S 13° 20.6', HP 55.91' ($SD \approx 0.2725 \times HP = 15.24'$).

Notes

- Accuracy 0.002' (HP 0.001'). Tables valid 1 Sep 2026 – 31 Dec 2027; later years with `c47_almanac_generator.py`.
- Not yet run on the C47: check RCL IND and DSE with the Sun example.

Using the results for Hc and Zn

- GHA and Dec at the sight time (UT = UTC + DUT1): SUNA (Sun), STAR (stars), ALM (Moon, planets).
- $LHA = GHA + \lambda$ (E +), mod 360.
- $Hc = \text{asin}(\sin \phi \sin \delta + \cos \phi \cos \delta \cos LHA)$.
- $Zn = \text{atan2}(-\cos \delta \sin LHA, \cos \phi \sin \delta - \sin \phi \cos \delta \cos LHA)$, mod 360.
- Intercept = $Ho - Hc$, toward Zn if positive. The C47 key $\rightarrow HcZ$ may do these steps directly.
- For Ho you still need index error, dip, refraction, SD ($\odot$ from R73, ☾ from ALM HP) and the almanac's planet corrections.

Python and NumWorks version

`nav.py` + `navdata.py` contain the same methods (SUNA, STAR, SUNRISE, PHAS) for a computer or a NumWorks. The menu asks for year, month, day and UT (hh:mm:ss) and computes the JD itself. Results are identical to the C47 programs (for example Sun 23 Sep 2026 12h: GHA 1° 54.4', Dec S 0° 11.6').

Register map

Program	Registers
SUNA	50-55, 60-67, 70-81
STAR	56-59, 68-69, 82-89
SUNRISE, HORZ	90-99
PHAS	41-48
SNAM	49
ALM	01-09, 10-33, 40-41
CHZ	91-92, 94-97
PTXB / PTXT	30-36
HALMV, ALMF, HALM	10-29, 37, 40-48 (work registers, results of SUNRISE/PHAS kept in 13-19)

C47 notes

- The C47 does not treat $\rightarrow \text{DEG}$ and flag 77 like the HP-42S; the programs avoid both (verified on the C47 with SUNA).
- Memory kept on the calculator: about 12 KB (programs + matrices) of about 246 KB free.

- Keep the printed Nautical Almanac as the primary reference; use the calculator as support and cross-check.